Mid Semester Examination-2 (Nov 2023)
Data Warehousing and Data Mining (ITPE77)

Time Allowed: 50 m
Max. Marks: 30

Note- **Attempt all questions.**

1 A database has five transactions. Let $min_sup = 60\%$ and $min_conf = 80\%$.

T ID	Items_bought
T100	{M, O, N, K, E, Y}
T200	{D, O, N, K, E, Y}
T300	{M, A, K, E}
T400	{M, U, C, K, Y}
T500	{C, O, K, I, E}

(a) Find all frequent itemsets using Apriori and FP-growth, respectively. Compare the efficiency of the two mining processes.

(b) List all the *strong* association rules (with support s and confidence c) matching the following metarule, where X is a variable representing customers, and $item_i$ denotes variables representing items (e.g., "A," "B,"):

$$\forall x \in transaction, buys(X, item_1) \wedge buys(X, item_2) \Rightarrow buys(X, item_3) \quad [s, c]$$

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2(a) Construct FP tree for the following transaction data set

TID	Items
1	{a,b}
2	{b,c,d}
3	{a,c,d,e}
4	{a,d,e}
5	{a,b,c}
6	{a,b,c,d}
7	{a}
8	{a,b,c}
9	{a,b,d}
10	{b,c,e}

5

Continue on back side ...

(b) Consider the following data set for a binary class problem.

A	B	Class Label
T	F	+
T	T	+
T	T	+
T	F	−
T	T	+
F	F	−
F	F	−
F	F	−
T	T	−
T	F	−

Calculate the information gain when splitting on A and B. Which attribute would the decision tree induction algorithm choose? 5

3 (a) Which are the two properties possessed by a rule set a rule based classifier?

(b) What is model overfitting in decision construction? Why does it occur?

(c) What does lattice of structure represent?

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